Lab 6. Analysis of variance with statsmodels

Analysis of variance (ANOVA) is a statistical method used to compare the means of two or more groups. It involves analyzing how much of the total variation in the data is due to differences between groups and how much is due to differences within groups.

During this mini-project, we will explore the `statsmodels` library and learn how to apply it through an example of statistical inference with ANOVA.

Theoretical Intruduction to ANOVA

Consider a situation in which we have three different teaching methods (A, B, C) and we want to check whether they differ in their impact on students' average performance. ANOVA makes it possible to compare all groups simultaneously. It is based on a linear model, and to describe it formally, let us assume that we have k groups, and each group has n_i observations, where $i \in \{1, \dots, k\}$.

The general model can be written as:

$$Y_{ij} = \mu_i + \epsilon_{ij}, \quad j = 1, \dots, n_i,\ j = 1, \dots, k,$$

where Y_{ij} denotes the j-th observation in the i-th group.

Additionally, the group mean μ_i can be expressed as:

$$\mu_i = \mu + \beta_i, \quad i = 1, \dots, k,$$

where μ is the overall mean, and β_i is the effect of the i-th group.

We also impose the constraint:

$$\beta_1 + \beta_2 + \cdots + \beta_k = 0.$$

Without such a constraint, the parameters μ and β_i would not be unique determined.

Analysis of variance is applied in many areas, for example:

- **Medicine**: Y could represent a patient's recovery time, and the groups could represent different treatment methods.
- **Agriculture**: Y could be crop yield, and the groups could represent environmental conditions such as lighting, irrigation, or fertilization.
- **Finance**: Y could represent creditworthiness, and the groups could correspond to factors such as education, occupation, marital status, or age group.
- **Genetics**: Y could be an organism's height, and the groups could correspond to genotypes determined by specific genes.

Task 1. (Ilustrative example)

Generate data and perform preliminary analysis.

Instructions:

- Generate three data arrays of size $n = 25$, each representing a different group (A, B, C). The data for each group should be sampled from a normal distribution with the same standard deviation $\sigma = 10$, but with different means:
  - Group A: $\mu_A = 65$;
  - Group B: $\mu_B = 70$;
  - Group C: $\mu_C = 75$.
- Store data in a DataFrame with columns:
  - `score` - simulated values
  - `group` - group labels ("A", "B", "C").

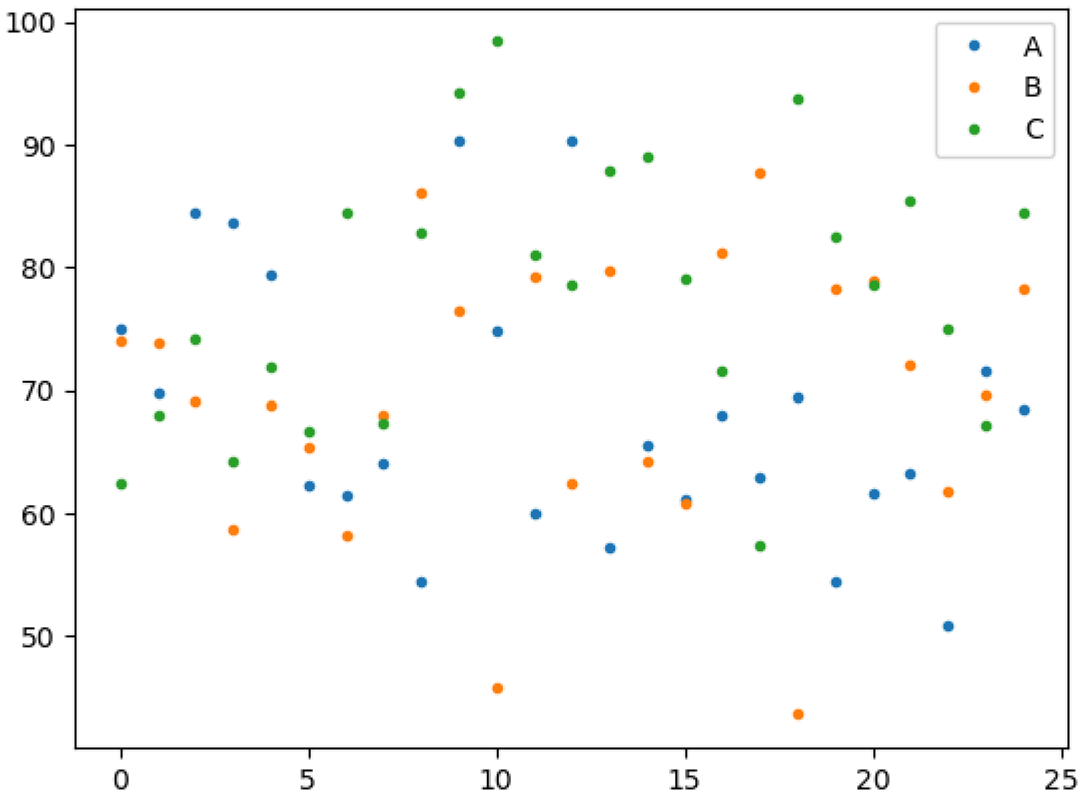
  For example, you can create three DataFrames and concatenate them.
- Compute the overall mean of all data and the mean of each group. Try `groupby`.
- Create a plot of the data, using a different color for each group.

In [12]:
```
import numpy as np
import pandas as pd
```

```
import matplotlib.pyplot as plt

n=25
sigma=10
A=np.random.normal(65, sigma, n)
B=np.random.normal(70, sigma, n)
C1=np.random.normal(75, sigma, n)

arr_A=pd.DataFrame([np.array(n*["A"]), A]).T
arr_B=pd.DataFrame([np.array(n*["B"]), B]).T
arr_C=pd.DataFrame([np.array(n*["C"]), C1]).T

df=pd.concat((arr_A, arr_B, arr_C))
df.columns=["group", "score"]
print(df)

print(df.groupby("group").mean())

for group_name, group_df in df.groupby("group"):
    plt.plot(group_df["score"].values, label=group_name, linestyle="", marker=".")
plt.legend()
```

```
  group      score
0      A  75.017203
1      A  69.799651
2      A  84.455575
3      A  83.580153
4      A   79.32201
..   ...        ...
20     C  78.564597
21     C  85.351938
22     C  74.918898
23     C  67.122212
24     C  84.506748

[75 rows x 2 columns]
           score
group
A      68.152447
B      69.676202
C       77.81873
```

Out[12]: <matplotlib.legend.Legend at 0x7510dd032050>



One-Way Analysis of Variance (ANOVA)

In a one-way ANOVA, we study the relationship between a **quantitative variable** Y (the response) and **one qualitative variable** (a factor) that has k levels (it is a k differnt groups).

We ask the question: *Does the mean value of Y differ significantly depending on the factor level?*

Hypotheses

The null hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k,$$

i.e, the mean of Y is the same for all groups (no effect of the factor).

The alternative hypothesis

$$H_1 : \exists_{i,j}\ \mu_i \neq \mu_j,$$

i.e, at least one group mean is different (the factor has an effect).

Equivalently, using the linear model with group effects β_i:

$$H_0 : \beta_1 = \beta_2 = \dots, \beta_k = 0 \quad \text{vs} \quad H_1 : \exists_{i,j}\ \beta_i \neq \beta_j.$$

The F-Test in ANOVA

To test these hypotheses, we use the **F-test**. If H_0 is true, $\bar{Y}_1, \dots, \bar{Y}_k$ should not differ much, where $\bar{Y}_i$ is the sample mean of group i:

$$\bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}.$$

It can be shown that $\bar{Y}_i$ is an unbiased estimator of μ_i.

Decomposition of Total Variability

The **total sum of squares (T_{SST})** measures the total variation of all data around the overall mean:

$$T_{SST} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(Y_{ij} - \bar{Y}\right)^2,$$

where

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^{k} \sum_{j=1}^{n_i} Y_{ij}$$

is an overall mean and $n = n_1 + \cdots + n_k$.

We can decompose T_{SST} as:

$$T_{SST} = T_{SSA} + T_{SSE},$$

where

- $T_{SSA} = \sum_{i=1}^{k} n_i \left(\bar{Y}_i - \bar{Y}\right)^2$ (sum of squares among groups),
- $T_{SSE} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(Y_{ij} - \bar{Y}_i\right)^2$ (sum of squares within groups).

It can be shown that

$$\frac{1}{\sigma^2} T_{SSA} \sim \chi^2(k-1),$$

and

$$\frac{1}{\sigma^2} T_{SSE} \sim \chi^2\left(n-k\right).$$

Therefore, the test statistic has a form

$$F = \frac{\frac{1}{k-1} T_{SSA}}{\frac{1}{n-k} T_{SSE}}$$

and follows an F **distribution** with $k-1$ and $n-k$ degrees of freedom when H_0 is true.

Then, we compute the p-value, which is the probability of obtaining a value equal to or more extreme than the observed test statistic

$$p = \mathrm{P}(F_{k-1,n-k} \geq F_{observed}).$$

If the resulting p**-value <** α **(significance level), we reject** H_0 and conclude that not all group means are equal.

Task 2. (Calculate the F statistic manually and using the statsmodels)

In this task, we use the dataset generated in Task 1 and calculate the F statistic manually and using the In this task, you will use the dataset from Task 1 to compute the F-statistic both by hand and using the statsmodels library.

Instructions:

1. Manual calculation:

   - From the dataset, compute:
     - T_{SSA} (between-group),
     - T_{SSE} (within-group),
     - T_{SST} (total),
     - verify that $T_{SST} = T_{SSA} + T_{SSE}$.
   - Use these values to manually calculate the F-statistic using the adequate formula. Recall $k =$ number of groups and $n =$ the total number of observations.

   - Find the corresponding p-value from the F-distribution. Tou can use `scipy.stats.f.sf` for this step.

2. Using statsmodels:

   - Fit the model with `ols` from `statsmodels.formula.api`:

   ```
   from statsmodels.formula.api import ols
   model = ols('score ~ C(group)', data=df).fit()
   ```

   Here, `score` is the dependent variable, and `C(group)` specifies that group is categorical.

   - Secondly, run an ANOVA on this model using:

   ```
   from statsmodels.stats.anova import anova_lm
   anova_results = anova_lm(model)
   print(anova_results)
   ```

   - Interpret the output: does the ANOVA test show significant differences between the group means?

In [59]:
```
from scipy.stats import f
k=3
N=3*n

Y_mean=1/N*sum(df["score"])
```

```
T_SST=sum((df["score"]-Y_mean)**2)
print(f"T_SST: {T_SST}")

group_means=df.groupby("group").mean()["score"]
T_SSA=n*(group_means["A"]-Y_mean)**2+n*(group_means["B"]-Y_mean)**2+n*(group_means["C"]-Y_mean)**2
print(f"T_SSA: {T_SSA}")

T_SSE=sum((A-group_means["A"])**2)+sum((B-group_means["B"])**2)+sum((C1-group_means["C"])**2)
print(f"T_SSE: {T_SSE}")

print(f"\nT_SSA+T_SSE={round(T_SSA,4)}+{round(T_SSE,4)}={round(T_SST,4)}=T_SST")


F=(1/(k-1))*T_SSA/(1/(N-k)*T_SSE)
print(f"F={F}")

p_value = f.sf(F, k-1, N-k)
print(f"p_value: {p_value}")
```

```
T_SST: 9936.065581834831
T_SSA: 1350.496929481304
T_SSE: 8585.568652353528

T_SSA+T_SSE=1350.4969+8585.5687=9936.0656=T_SST
F=5.662745408016686
p_value: 0.005199554971264964
```

In [14]:

```
import scipy, statsmodels
from statsmodels.formula.api import ols
from statsmodels.stats.anova import anova_lm

df['score']=pd.to_numeric(df['score'])
model=ols('score ~ C(group)', data=df).fit()
anova_results = anova_lm(model)
print(anova_results)
```

```
            df       sum_sq     mean_sq         F  PR(>F)
C(group)   2.0  1350.496929  675.248465  5.662745  0.0052
Residual  72.0  8585.568652  119.244009       NaN     NaN
```

Our p_value is way smaller than 0.05, so we reject H_0 - it means that not all of means are equal.

Assumptions of ANOVA

Analysis of Variance requires following conditions to be satisfied:

1. **Normality**: the residuals are normally distributed
2. **Homoscedasticity**: all groups have equal variances.
3. **Independence**: observations are independent.

In the following tasks, we consider the new dataset `crop.data.csv`, which was generated for teaching purposes. It contains 97 observations of crop yield under different experimental conditions.

The dataset includes the following variables:

- `fertilizer` — a categorical factor indicating the type or mix of fertilizer applied, with three levels (1, 2, 3).
- `density` — a categorical factor denoting planting density, with three levels: low (1), medium (2), and high (3).
- `block` — a factor representing the experimental plot or block, with four possible values (1–4).
- `yield` — a continuous numeric variable measuring the crop yield, which serves as the dependent variable.

In [88]:
```
df_crop = pd.read_csv('crop.data.csv')
df_crop.head()
```

Out[88]:

	density	block	fertilizer	Yield
0	1	1	1	177.228692
1	2	2	1	177.550041
2	1	3	1	176.408462
3	2	4	1	177.703625
4	1	1	1	177.125486

Task 3. (Q–Q plot and histogram of residuals)

We can use a **Q–Q plot** and a **histogram of residuals** to visually check whether the residuals are normally distributed. If the points in the Q-Q plot follow the straight line and the histogram looks approximately normal, we can assume residuals are normally distributed.

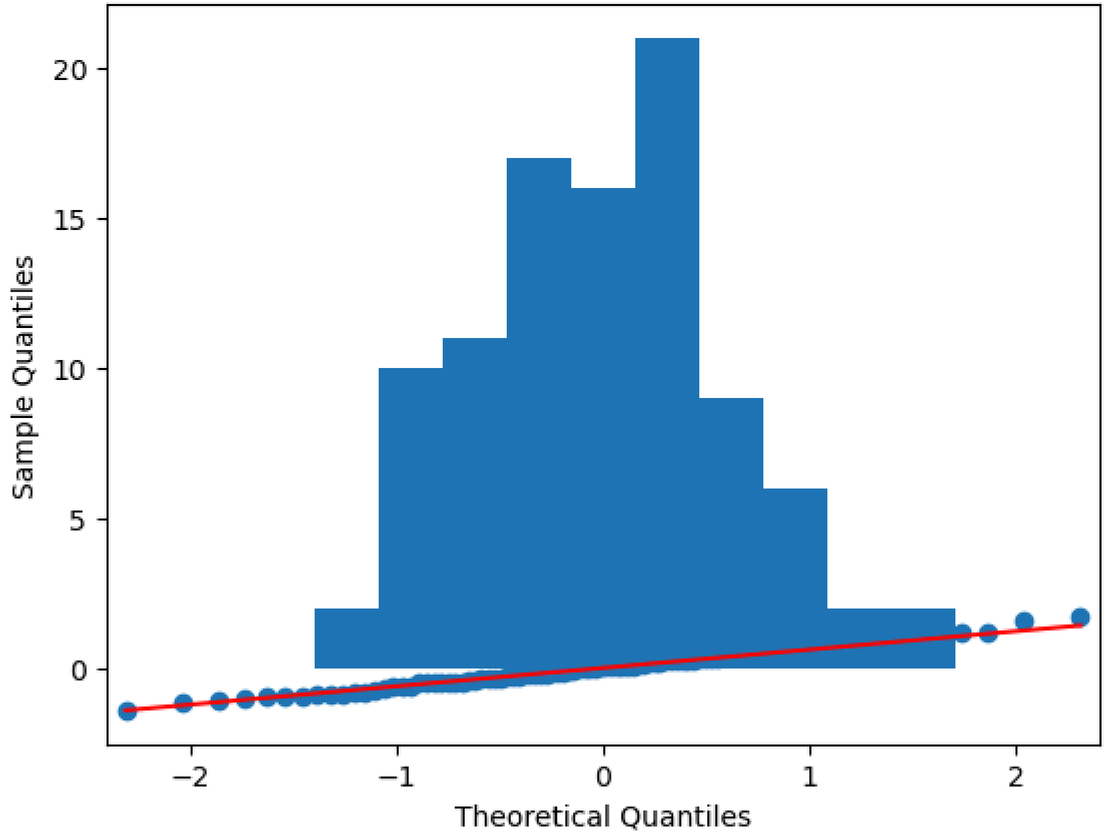
- Fit an ANOVA model and check the effect of `fertilizer` on `Yield`. Extract the residuals using `model.resid`.
- Use `sm.qqplot(residuals, line='s')` form library `statsmodels.api` to check if residuals follow a normal distribution.
- Use `matplotlib.pyplot.hist(residuals)` to visualize distribution and detect skewness or outliers.

In [89]:

```
import statsmodels.api as sm
import matplotlib
model2=ols('Yield ~ C(fertilizer)', data=df_crop).fit()
results_crop = anova_lm(model2)
print(results_crop)

residuals=model2.resid
sm.qqplot(residuals, line="s")
matplotlib.pyplot.hist(residuals)
```

```
                 df     sum_sq   mean_sq         F  PR(>F)
C(fertilizer)   2.0   6.068047  3.034023  7.862752  0.0007
Residual       93.0  35.886186  0.385873       NaN     NaN
```

Out[89]:
```
(array([ 2., 10., 11., 17., 16., 21.,  9.,  6.,  2.,  2.]),
 array([-1.39620473, -1.08611135, -0.77601796, -0.46592458, -0.15583119,
         0.1542622 ,  0.46435558,  0.77444897,  1.08454235,  1.39463574,
         1.70472913]),
 <BarContainer object of 10 artists>)
```



Task 4. (Shapiro-Wilk test)

The **Shapiro–Wilk test** formally checks if residuals follow a normal distribution. The hypotheses in the Shapiro–Wilk test are:

$$H_0 : \text{Residuals are normally distributed} \qquad H_1 : \text{Residuals are not normally distributed.}$$

- Apply the function `shapiro` from `scipy.stats` to test whether the residuals of the ANOVA model are normally distributed.

- Interpretation of the test:
  - If the **p-value > 0.05**, we do not reject H_0. This means the residuals can be considered normally distributed.
  - If the **p-value ≤ 0.05**, we reject H_0. This means the residuals deviate significantly from normality.

In [90]:
```
scipy.stats.shapiro(residuals)
```

Out[90]:
```
ShapiroResult(statistic=0.9908844232559204, pvalue=0.759391188621521)
```

As p-value is way bigger than 0.05, we assume that the residuals are normally distributed.

Task 5. (Levene's test)

We test whether the variances in groups are equal. For that, we can use **Levene's test**. The hypotheses in the Levene's test are:

$$H_0 : \sigma_1^2 = \sigma_2^2 = \ldots, \sigma_k^2 \text{ equal variances across groups} \qquad H_1 : \exists_{i,j} \sigma_i^2 \neq \sigma_j^2 \text{ at least two variances differ.}$$

- Apply the function `levene` from `scipy.stats` to test whether the variances of the groups are equal.
- Syntax of the function:

  ```
  scipy.stats.levene(sample1, sample2, ..., center='median')
  ```
  where `sample1`, `sample2`, ... are arrays of observations from each group.
- Interprate:
  - If **p-value > 0.05**, do not reject H_0. The assumption of equal variances is satisfied.
  - If **p-value ≤ 0.05**, reject H_0. The assumption of equal variances is violated, and ANOVA results may not be reliable

In [91]:
```
sample1=df_crop[df_crop["fertilizer"]==1]["Yield"]
sample2=df_crop[df_crop["fertilizer"]==2]["Yield"]
sample3=df_crop[df_crop["fertilizer"]==3]["Yield"]

scipy.stats.levene(sample1, sample2, sample3, center="median")
```

Out[91]:
```
LeveneResult(statistic=0.8471814796777352, pvalue=0.43190199882090996)
```

As p-value is way bigger than 0.05, we assume that the variances among groups are equal.

Task 6. (Durbin-Watson test)

Independence of observations cannot be fully verified using plots; it is usually ensured by the experimental design. However, we can check for autocorrelation in residuals with the **Durbin–Watson test**.

- Using the dataset from Task 3, apply the function `durbin_watson` from the library `statsmodels.stats.stattools` to evaluate the independence of the residuals in your ANOVA model.
- The hypotheses are:

$$H_0: \text{ no correlation of residuals } \rho = 0 \qquad H_1: \text{residuals are correlated } \rho \neq 0.$$

- Interpretation of the Durbin–Watson statistic:
  - A value **close to** 2 suggests no autocorrelation.
  - Values **below** 1 or **above** 3 indicate strong autocorrelation.

In [36]:
```
statsmodels.stats.stattools.durbin_watson(residuals)
```

Out[36]:
```
2.3651074322841783
```

The result is close to 2, so we don't reject H_0.

Post-Hoc Tests in ANOVA

After performing ANOVA, we often want to know **which groups differ significantly**. Post-hoc tests are designed for this purpose.

Problem of Multiple Testing

When comparing more than two groups, the number of pairwise comparisons grows quickly. For example:

- With 3 groups, we have 3 pairwise comparisons.
- With 10 groups, we need 45 comparisons.

If we simply performed multiple t-tests (The t-test is a statistical test used to compare the means of two groups), the probability of making a Type I error (false positive) increases. Recall, that Type I error is rejecting true null hypothesis.

Example: With 3 hypotheses tested at $\alpha = 0.05$, the probability of at least one false positive is:

$$P(\text{at least one false positive}) = 1 - (1 - 0.05)^3 \approx 0.1426$$

That's about 14%, even if all null hypotheses are true. With 10 groups, the probability rises above 90%.

Post-hoc tests (like Tukey's HSD) solve this by controlling the overall Type I error rate.

Tukey's HSD Test

The **Tukey's Honest Significant Difference (HSD)** test compares all possible pairs of group means while adjusting for multiple testing.

Based on the means with differenc groups $\bar{Y}_1, \ldots, \bar{Y}_k$ we compue the test statistic:

$$t_{ij} = \frac{\bar{Y}_i - \bar{Y}_j}{\sqrt{s^2 \frac{1}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}},$$

where:

- n_i, n_j are the groups sizes,
- $s^2 = \frac{T_{SSE}}{n-k}$ - mean squeare error from ANOVA,
- k - number of groups,
- n - total numbers of observations.

We calculate $\frac{k(k-1)}{2}$ such test statistics and test hypotheses:

$$H_0 : \mu_i = \mu_j \quad \text{(no difference between groups)} \qquad H_1 : \mu_i \neq \mu_j \quad \text{(means differ).}$$

The critical values come from the Tukey distribution. The p-values can be computes using the below formula:

$$p = \mathrm{P}(T_{k,n-k} \geq |t_{ij}|)$$

If the p-value is less than the significance level (e.g., 0.05), we reject H_0.

Task 7. (Tukey's HSD test)

Perform Tukey's HSD test to compate all possible pairs of group means.

- Fit a one-way ANOVA model to test whether the factor `fertilizer` has a significant effect on `Yield`, and interpret the results.
- If the ANOVA result is significant, perform **Tukey's HSD test** to identify which pairs of group means differ. You can use `pairwise_tukeyhsd` from `statsmodels.stats.multicomp`.
- Manually calculate the p-value for the comparison between **group 2 and group 3** and compare it with the result from the built-in function. To do this, use the fitted ANOVA model to extract the necessary values, and compute the Tukey distribution with `psturng` from `statsmodels.stats.libqsturng`.
- Finally, visualize the results of Tukey's HSD test. You can use the built-in method `plot_simultaneous()` on the `pairwise_tukeyhsd` object to produce a graphical summary.

In [37]:
```
#we already did the first one in Task 3:
print(results_crop)
```

```
                 df     sum_sq   mean_sq         F  PR(>F)
C(fertilizer)   2.0   6.068047  3.034023  7.862752  0.0007
Residual       93.0  35.886186  0.385873       NaN     NaN
```

We may see that p-value is smaller than 0.05, which means, that we reject the H_0 hypothesis that all of the groups have the same mean. We are going to find the pairs that don't.

In [65]:
```
crop_yield=df_crop["Yield"]
fert=df_crop["fertilizer"]
tukey=statsmodels.stats.multicomp.pairwise_tukeyhsd(endog=crop_yield, groups=fert)
print(tukey)
```

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
===================================================
group1 group2 meandiff p-adj   lower  upper  reject
---------------------------------------------------
     1      2   0.1762 0.4955 -0.1937 0.5461  False
     1      3   0.5991 0.0006  0.2292  0.969   True
     2      3    0.423 0.0209  0.0531 0.7928   True
---------------------------------------------------
```

Looks like group 1 and 2 have the same means (or more like we don't reject the hypothesis they do), but group 3 differs. Let's compare groups 2 and 3:

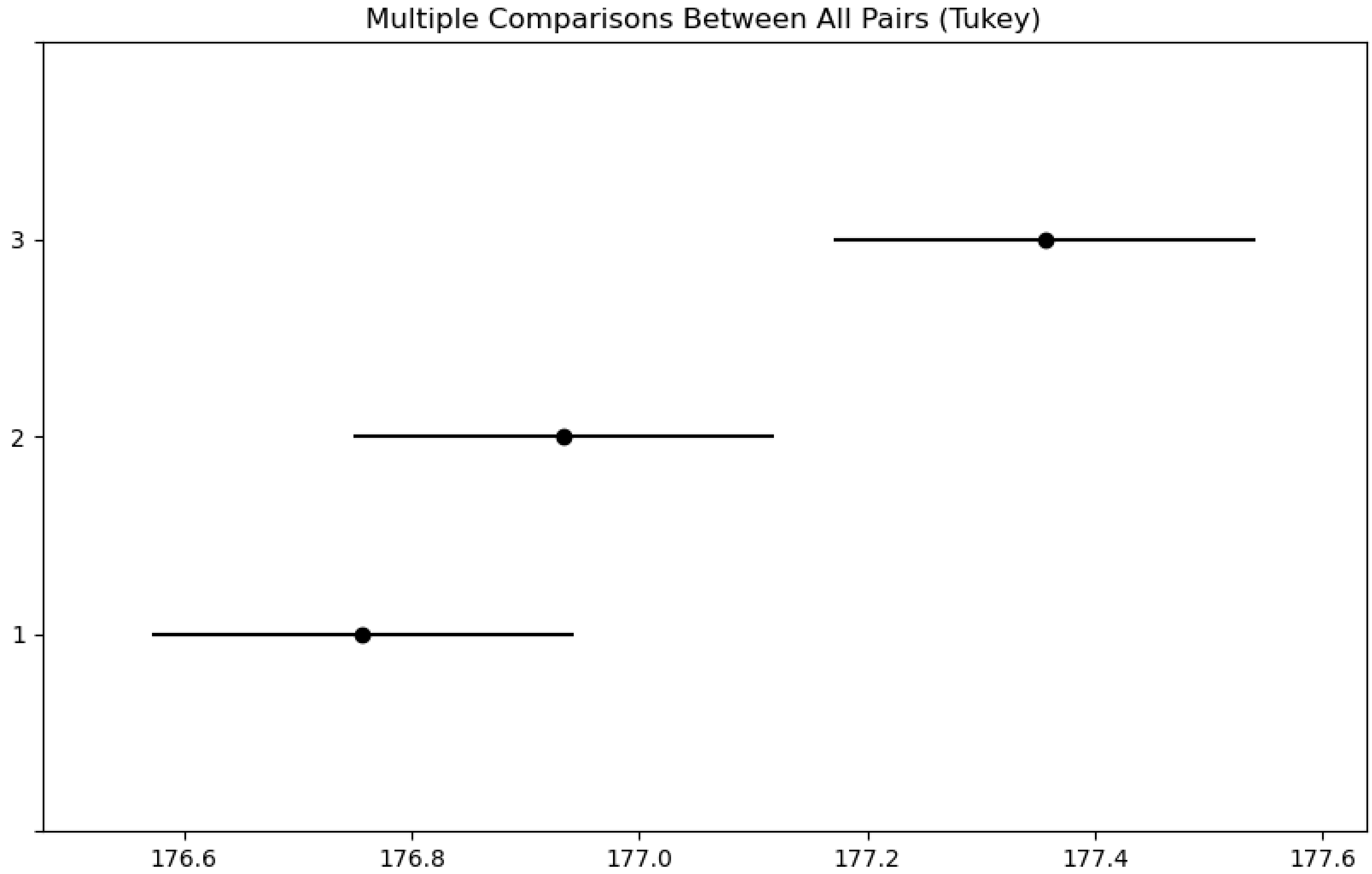
In [64]:
```
from statsmodels.stats.libqsturng import psturng

n_2=len(sample2)
n_3=len(sample3)
Y_2_mean=np.mean(sample2)
Y_3_mean=np.mean(sample3)

n_cr=len(df_crop)
k_cr=3
T_SSE_cr=sum((sample2-Y_2_mean)**2)+sum((sample3-Y_3_mean)**2)

t_23=(Y_2_mean-Y_3_mean)/np.sqrt(T_SSE_cr/(n_cr-k_cr)*0.5*(1/n_2+1/n_3))
print(psturng(abs(t_23), k_cr, n_cr-k_cr))
```

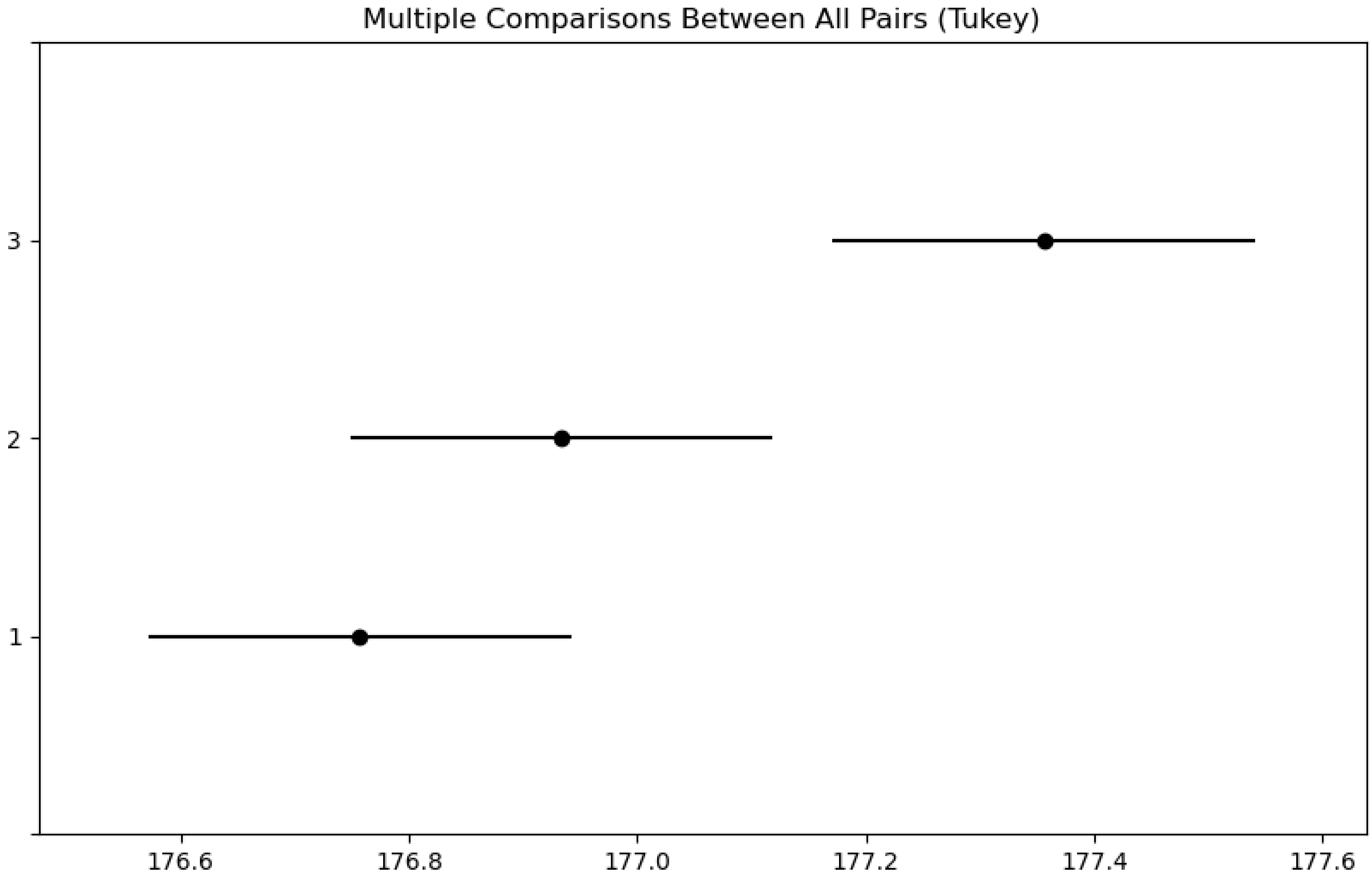
```
[0.00185019]
```

The p-value is smaller than 0.05, so we reject H_0 - groups 2 and 3 have indeed different means.

In [70]:
```
tukey.plot_simultaneous()
```

Out[70]:


Multiple Comparisons Between All Pairs (Tukey)
3
2
1
176.6
176.8
177.0
177.2
177.4
177.6



The graphs confirm what we expected - group 1 and 2 may have the same mean (we didn't reject H_0), but group 3 definitely does not have the same mean with either of them.

Two-Way (Multi-Way) ANOVA

In **one-way ANOVA**, we studied the mean values of a dependent variable Y across different groups of **one categorical variable**.

In **two-way (or multi-way) ANOVA**, we study the mean values of Y across combinations of **two (or more) categorical variables.**

We distinguish two types of models:

- Additive model (no interactions),
- Model with interactions.

If we want to test whether the mean of Y differs across groups defined by multiple categorical variables, we could run separate one-way ANOVAs for each explanatory variable. However, this approach **ignores possible dependence between variables.**

A better approach is to include multiple categorical variables in **one model** and test differences in expected values of Y across groups. Note that including too many variables may reduce model interpretability.

Additive Model (No Interaction)

In the **additive two-way ANOVA**, we assume **no interaction between factors**, i.e., each factor affects the dependent variable independently of the other factor.

The additive model can be written as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk},$$

where:

- Y_{ijk} dependent variable for observation k in group i of factor A and group j of factor B,
- μ overall mean,
- α_i effect of the i-th level of factor A,
- β_j effect of the j-th level of factor B,
- ϵ_{ijk} error term, assumed i.i.d. $N(0, \sigma^2)$.

Constraints:

$$\sum_i \alpha_i = 0 \text{ and } \sum_j \beta_j = 0.$$

Hypotheses: For factor A:

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a \qquad H_1 : \alpha_i \neq 0 \text{ for at least one } i,$$

For factor B:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b \qquad H_1 : \beta_j \neq 0 \text{ for at least one } j.$$

Task 8. (Two-way ANOVA)

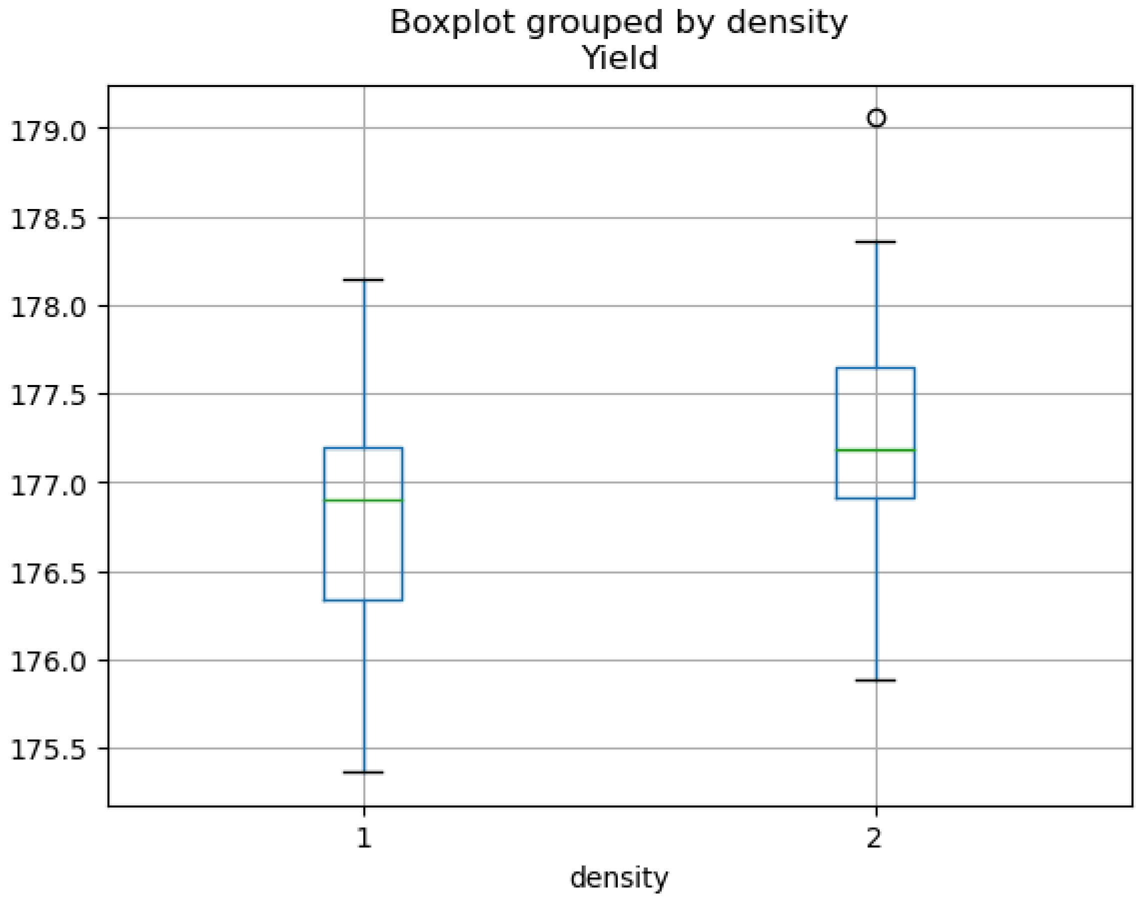
Investigate whether the variable `Yield` depends on two categorical factors: `density` and `fertilizer`.

- Create two boxplots:
  - one showing `Yield` by `density`,
  - another showing `Yield` by `fertilizer`.
- Fit an additive two-way ANOVA model using the formula:

  ```
  model = ols('Yield ~ C(density) + C(fertilizer)', data=df).fit()
  ```
- If either factor is significant, perform post-hoc comparisons to determine which levels differ, analyzing each factor separately.

In [76]:
```
df_crop.boxplot(column="Yield", by="density")
df_crop.boxplot(column="Yield", by="fertilizer")

model_two_way = ols('Yield ~ C(density) + C(fertilizer)', data=df_crop).fit()
print(anova_lm(model_two_way))
```

```
                 df     sum_sq   mean_sq          F    PR(>F)
C(density)      1.0   5.121681  5.121681  15.316179  0.000174
C(fertilizer)   2.0   6.068047  3.034023   9.073123  0.000253
Residual       92.0  30.764505  0.334397        NaN       NaN
```


Boxplot grouped by density
Yield
179.0
178.5
178.0
177.5
177.0
176.5
176.0
175.5
1
2
density

Both factors are significant, so we should make two analysis, but we already did everything for fertilizer above, and density has only two groups - they must be the ones that differ... Tukey won't work here, we can only do one-way ANOVA, which will only show us the same thing (p_value<0.05).

(if you require something more here, let me know and I'll correct it, but right now I think nothing more can be done here)

In [79]:
```
model_dens=ols('Yield ~ C(density)', data=df_crop).fit()
results_dens = anova_lm(model_dens)
print(results_dens)
```

```
              df     sum_sq   mean_sq          F    PR(>F)
C(density)   1.0   5.121681  5.121681  13.070993  0.000485
Residual    94.0  36.832552  0.391836        NaN       NaN
```

Two-Way ANOVA with Interaction

Sometimes two factors do **not act independently** on the mean of Y. There may be an interaction effect, meaning that the effect of one factor depends on the level of the other factor.

The two-way ANOVA model with interaction:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk},$$

where

- α_i - effect of factor A,
- β_j - effecto of factor B,
- $(\alpha\beta)_{ij}$ - interation effect between A and B,
- ϵ_{ijk} - random error.

Hypothesis to test:

1. Factor A effect:

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a$$

2. Factor B effect:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_a$$

3. Interaction effect:

$$H_0 : (\alpha\beta)_{11} = (\alpha\beta)_{12} = \cdots = (\alpha\beta)_{ab} = 0$$

Visualize the interactions

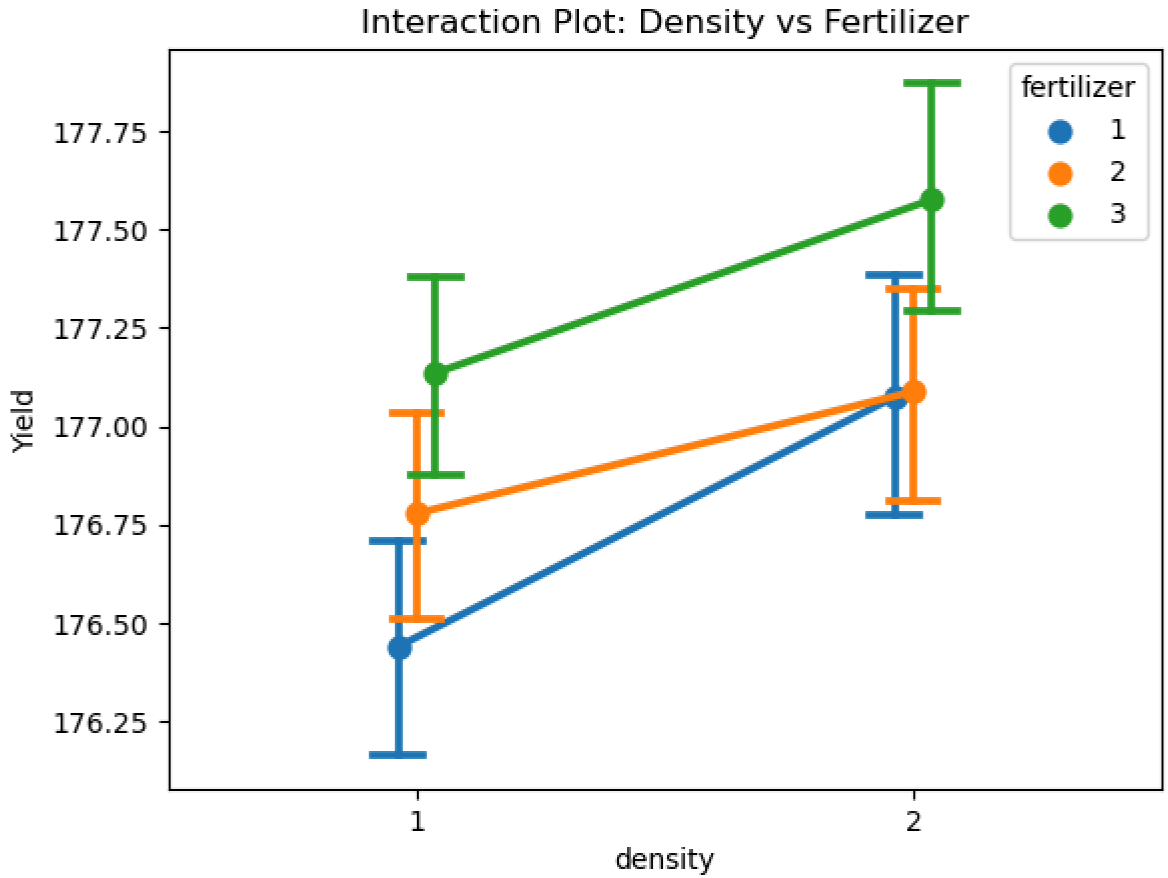
Fist, we can visualize interactions using

```
sns.pointplot(x='density', y='Yield', hue='fertilizer', data=df, dodge=True, capsize=0.1)
plt.title('Interaction Plot: Density vs Fertilizer')
```

- Parallel lines → additive effect, no interaction.
- Non-parallel lines → significant interaction, the effect of one factor depends on the other.

In [83]:
```
import seaborn as sns
sns.pointplot(x='density', y='Yield', hue='fertilizer', data=df_crop, dodge=True, capsize=0.1)
plt.title('Interaction Plot: Density vs Fertilizer')
```

Out[83]:
```
Text(0.5, 1.0, 'Interaction Plot: Density vs Fertilizer')
```



Lines are not parallel, but they don't differ that much - we'll check significance of that difference in a moment.

Post-Hoc Test for Interaction Model

In **multi-factor ANOVA**, we can perform **Tukey's Honest Significant Difference (HSD) test** to check which groups differ significantly.

For the interaction model, we compare:

- Main effects of factor A (density)
- Main effects of factor B (fertilizer)

- Interaction combinations (density × fertilizer)

Task 9. (Two-way ANOVA with interactions)

Investigate whether the variable `Yield` depends on the interaction between two categorical factors: `density` and `fertilizer`.

- Fit a two-way ANOVA model with interaction using the formula:

  ```
  model = ols('Yield ~ C(density) * C(fertilizer)', data=df).fit()
  ```
- Interpret the results of the model, paying special attention to the interaction term.

- If the interaction is significant, perform post-hoc comparisons to identify which specific combinations of factor levels differ. To do this, create a column in the DataFrame representing the interaction factor (e.g., `"density-fertilizer"`), and use it as a single categorical variable for post-hoc testing.

In [84]:
```
model_inter = ols('Yield ~ C(density) * C(fertilizer)', data=df_crop).fit()
anova_lm(model_inter)
```

Out[84]:

	df	sum_sq	mean_sq	F	PR(>F)
C(density)	1.0	5.121681	5.121681	15.194517	0.000186
C(fertilizer)	2.0	6.068047	3.034023	9.001052	0.000273
C(density):C(fertilizer)	2.0	0.427818	0.213909	0.634605	0.532500
Residual	90.0	30.336687	0.337074	NaN	NaN

We can see that interaction is not significant (p_value>>0.05) - it means that fertilizer effects were consistent across densities, which generally makes sense, looking back to the upper graph with three lines. There are differences, as I said before, but I suppose they aren't statistically significant.